

Fuzzy PETS

Protocol Implementation

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 $(pp^*, D) \leftarrow \Sigma^*.Setup(sp^*, F_{k_{PRF}}, m)$

```

1 : (spPDS, spLSH) ← sp* test_fuzzypsi_parts/cli_args.inc.cpp/parse_cli_args (166)
2 : (H, L, K) ← spLSH
3 : {Hℓ,k}ℓ∈[L],k∈[K] ← $ H lsh/lsh_e2lsh.h/E2LSH::E2LSH (13)
4 : (pp, D) ← Σ.Setup(spPDS, FkPRF, m · L) cuckoo_filter.h/CuckooFilter::CuckooFilter (90)
5 : pp* ← (pp, {Hℓ,k}k∈[K],ℓ∈[L])
6 : return (pp*, D)

```

 $(locs, info) \leftarrow \Sigma^*.GetInfo(\{F_{k_{PRF}}(u)\}_{u \in \mathbf{u}})$

```

1 : locs, info ← ⊥ℓ
2 : for ℓ ∈ [L] do Compute LSH keys, call PRF/OPRF and extract bucket keys and fingerprints:
   test_fuzzypsi_parts/cuckoo_prf_keys.inc.cpp/cuckoo_prf_encoded_key_batches_range (40)
3 :   locs[ℓ], info[ℓ] ← Σ.GetInfo(FkPRF(u[ℓ]))
4 : return (locs, info)

```

Server-side (setup phase):

Do the actual PRF:

opr/opr_eval.inc.cpp/opr_server_prf_blocks (78)

Client-side (online phase):

Do the actual OPRF:

opr/opr_eval.inc.cpp/opr_blocks (60)

$$\mathbf{u} \leftarrow \Sigma^*. \text{GetFuzzyVals}(x)$$
$$1: (_, \{H_{\ell,k}\}_{\ell \in [L], k \in [K]}) \leftarrow \text{pp}^*$$
$$2: \quad \mathbf{u} \leftarrow \perp^\ell$$
3 : **for** $\ell \in [L]$ **do**

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4: // Compute hash of “fuzzy” value.
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5: $\mathbf{u}[\ell] \leftarrow H(H_{\ell,1}(\mathbf{x}), \dots, H_{\ell,K}(\mathbf{x}))$ lsh/lsh_e2lsh.h/E2LSH::table_keys (56)

```

6 :   return u

```

$$D \leftarrow \Sigma^*.\text{Insert}(D, \text{locs}, \text{info})$$
1 : **for** $\ell \in [L]$ **do**2: $D \leftarrow \Sigma.\text{Insert}(D, \text{locs}[\ell], \text{info}[\ell])$

```
) 3: return D
```

resps $\leftarrow \Sigma^*.Query(D, locs)$

1: **resps** $\leftarrow \perp^\ell$
2: **for** $\ell \in [L]$ **do**
3: **resps** $[\ell] \leftarrow \Sigma.Query(D, locs[\ell])$
4: **return** **resps**

b $\leftarrow \Sigma^*.CheckMemb(resps, info)$

1: **for** $\ell \in [L]$ **do**
2: **b** $_\ell \leftarrow \Sigma.CheckMemb(resps[\ell], info[\ell])$
3: **return** $\bigvee_{\ell \in [L]} b_\ell$

collect candidate CF buckets (OR across tables)

pir/pir_membership.inc.cpp/cuckoo_membership_pir_batch_lookups (79)

retrieve bucket rows through BatchPIR

pir/batchpir/batchpir_row_reader.cpp/BatchPirRowReader::read_rows (236)

Check fingerprint of a PIR bucket:

cuckoo_filter.h/CuckooFilter::row_contains_fingerprint (230)

Server Offline Phase

Client C

1 :
2 :
3 :
4 :
5 :
6 :
7 :

Server S

```

 $k_{\text{PRF}} \leftarrow \{0, 1\}^\lambda$ 
 $(D, \text{pp}^*) \leftarrow \Sigma^*. \text{Setup}(\text{sp}^*, F_{k_{\text{PRF}}}, |Y|)$ 
// Insert values into fuzzy PDS
for  $y \in Y$  do
     $\mathbf{u} \leftarrow \Sigma^*. \text{GetFuzzyVals}(y)$ 
     $(\text{locs}, \text{info}) \leftarrow \Sigma^*. \text{GetInfo}(\{F_{k_{\text{PRF}}}(\mathbf{u})\}_{\mathbf{u} \in \mathbf{u}})$ 
     $D \leftarrow \Sigma^*. \text{Insert}(D, \text{locs}, \text{info})$ 
    
```

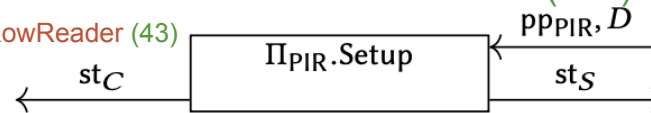
✓ already covered
in previous slides

PIR Offline Phase

$\text{PP}_{\text{PIR}} \leftarrow \Pi_{\text{PIR}}. \text{GenPP}(1^\lambda, |D|)$

pir/batchpir/batchpir_row_reader.cpp/BatchPirRowReader::BatchPirRowReader (167)

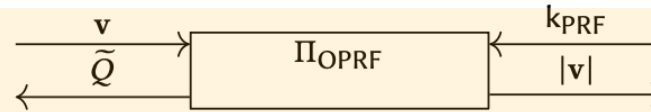
pir/checklist/checklist_readers.inc.cpp/ChecklistPirRowReader::ChecklistPirRowReader (43)



Online Phase

```

10 :  $I \leftarrow \emptyset; \text{resps} \leftarrow \perp^\ell$ 
11 : for  $x \in X$  do
12 :      $\mathbf{v} \leftarrow \Sigma^*. \text{GetFuzzyVals}(x)$  lsh/lsh_e2lsh.h/E2LSH::table_keys (56)
13 :     // Get OPRF output of fuzzy values
    
```



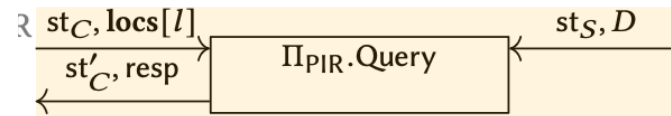
OPRF Call:
opr/opr_eval.inc.cpp/opr_blocks (60)

OPRF:
opr/droidcrypto/droidCrypto/psi/opr/opr.cpp/
doGCAES_OPRF_Blocks_Stats (144)

```

14 :      $\text{locs}, \text{info} \leftarrow \Sigma^*. \text{GetInfo}(\tilde{Q})$  test_fuzzypsi_parts/cuckoo_prf_keys.inc.cpp/
15 :     for  $\ell \in [L]$  do cuckoo_prf_encoded_key_batches_range (40)
    
```

pir/batchpir/batchpir_row_reader.cpp/
BatchPirRowReader::read_rows (236)



pir/checklist/checklist_readers.inc.cpp/
ChecklistPirRowReader::read_rows (68)

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17 :      $\text{resps}[\ell] \leftarrow \text{resp}$ 
18 :     if  $\Sigma^*. \text{CheckMemb}(\text{resps}, \text{info})$  do cuckoo_filter.h/CuckooFilter::row_contains_fingerprint (230) cuckoo_filter.h/CuckooFilter::lookup_encoded (116)
19 :          $I \leftarrow \bigcup \{x\}$  Calculate results, calculate FP/FN and etc. stats:
20 :         return I test_fuzzypsi_parts/
                experiment_runner.inc.cpp/fuzzypsi (198)
    
```